Introduction: Knowledge Representation & Reasoning

Dr. Yosser ATASSI | Informatics Engineering – Artificial Intelligence

Course Outline

1. Introduction & the role of logic in AI
2. Propositional logic
3. Reasoning in propositional logic
4. First-order logic: representation
5. Reasoning in first-order logic
6. Description logics & the Semantic Web
7. Non-monotonic logics
8. Temporal logic
9. Uncertainty & qualitative reasoning

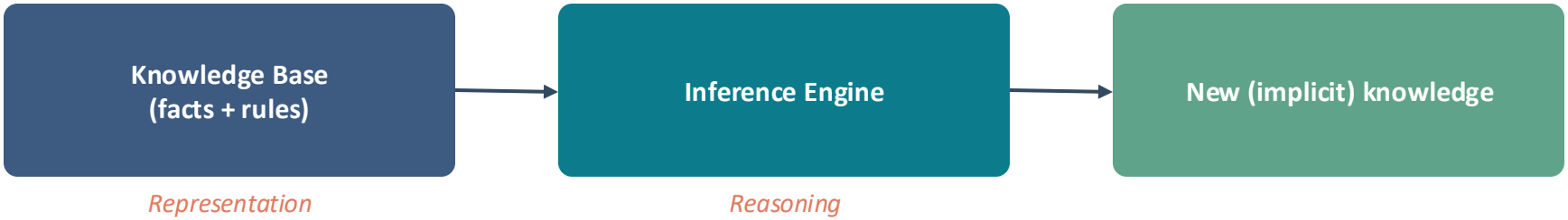
Today: topic 1 – why we need knowledge representation and how paradigms differ.

Today's Roadmap & Learning Objectives



- **Define:** knowledge representation and reasoning, and why the choice matters.
- **Write:** facts and rules in propositional logic and FOL.
- **Model:** knowledge with semantic networks, conceptual graphs and frames.
- **Compare:** symbolic, statistical and sub-symbolic approaches.
- **Justify:** why neuro-symbolic hybrids are an active research area.

What is Knowledge Representation & Reasoning?



Example: Socrates

Facts: "Socrates is a man."
"All men are mortal."
Inferred: "Therefore, Socrates is mortal."

In symbols

Man(Socrates)
$\forall x$ Man(x) $\rightarrow$ Mortal(x)
$\Longrightarrow$ Mortal(Socrates)

The Benefits of Using Logic

~~Explainability~~

The system can show the exact steps behind a conclusion:
"I concluded the murderer was the butler because…"

Precision

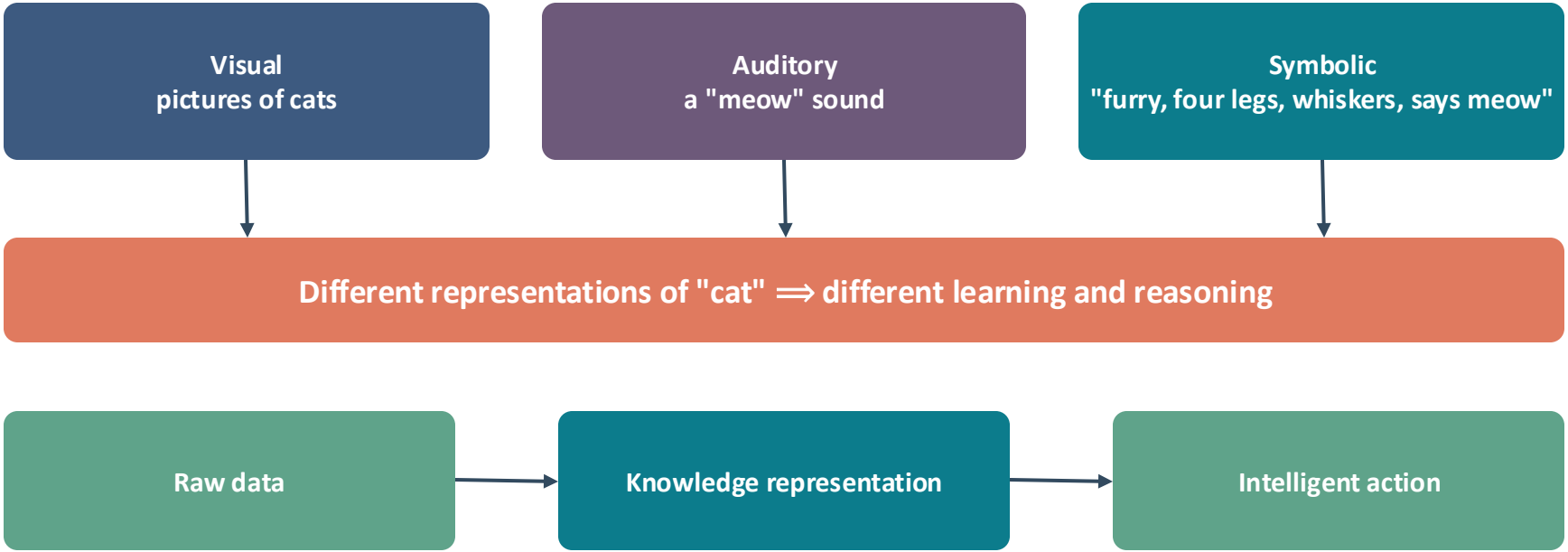
No ambiguity: a statement is either true or false within the model.

Generality

The same reasoning engine works for many problems, just by changing the knowledge base.

Applications: expert systems, legal reasoning, verification, the Semantic Web.

Why Representation Matters



KR is the bridge between raw data and intelligent action.

Core Goal of KR and the Three Paradigms

- **Facilitate inference:** derive new facts from known facts.
- **Be computationally efficient:** a machine can work with it effectively.

1. Symbolic

Logic, rules, networks, frames.
Human-readable, explainable.

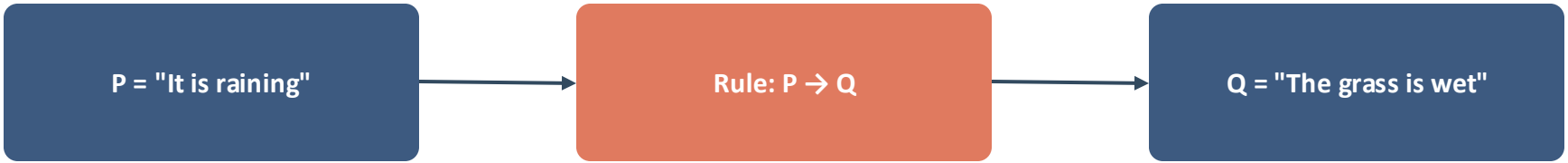
2. Statistical

Probabilities, Bayesian networks.
Handles uncertainty.

3. Sub-symbolic

Neural networks, weights.
Learns from data.

Method 1: Symbolic Representation (Propositional Logic)



- **Core idea:** the world can be described with logic and rules.
- **Propositional logic:** facts that are either True or False.
- If we know P is True, we infer Q is True (Modus Ponens).

First-Order Logic (Predicate Logic)

Example

Cat(Tom)
Mouse(Jerry)
Chases(Tom, Jerry) // Tom chases Jerry
∀x ∀y (Cat(x) ∧ Mouse(y) → Chases(x, y))

Ingredients

Predicates: properties and relations
Quantifiers: ∀ (for all), ∃ (exists)
Connectives: ∧ ∨ ¬ →

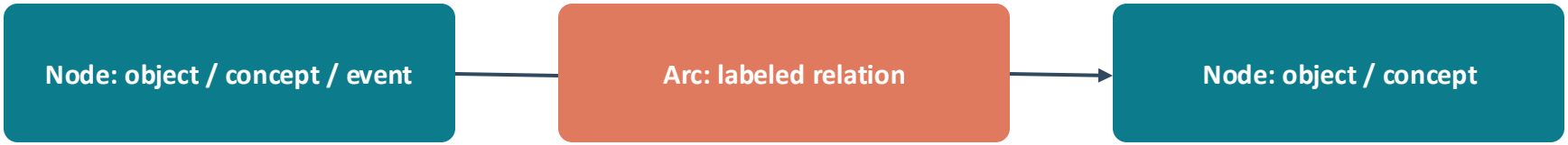
- **Pros:** explainable, precise, excellent for legal reasoning and expert systems.
- **Cons:** struggles with ambiguous, incomplete or perceptual data; rules are hand-crafted.

Worked Example: Propositional Logic vs FOL

Statement	Propositional logic	First-order logic
Tom is a cat	TomIsCat	Cat(Tom)
Jerry is a mouse	JerryIsMouse	Mouse(Jerry)
Tom chases Jerry	TomChasesJerry	Chases(Tom, Jerry)
Every cat chases every mouse	One rule per (cat, mouse) pair	$\forall x \forall y$ (Cat(x) $\wedge$ Mouse(y) $\rightarrow$ Chases(x,y))

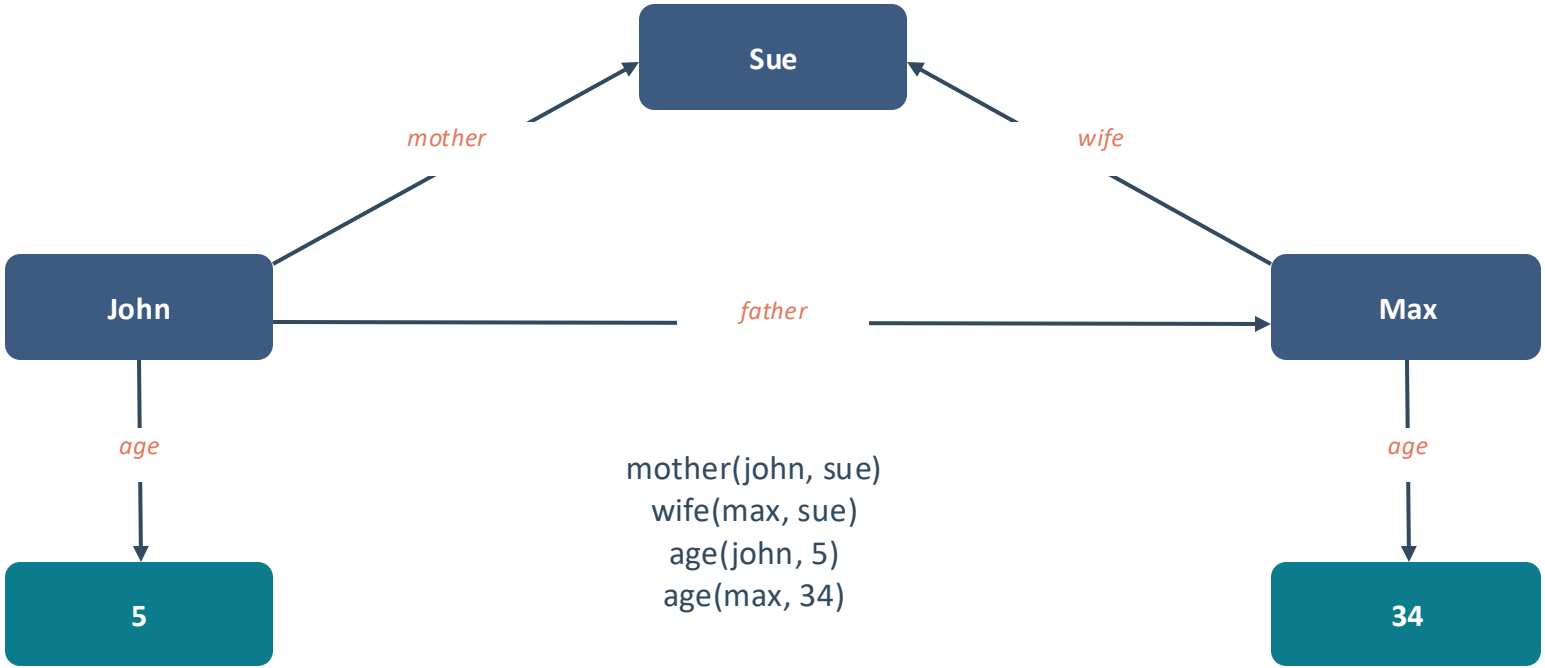
FOL = individuals + properties + generalizations in one rule.

Semantic Networks



- **A simple scheme:** a graph of labeled nodes and labeled directed arcs.
- **Nodes:** objects, concepts, events.
- **Arcs:** relationships between nodes.
- The graphical depiction is a big reason for their popularity.

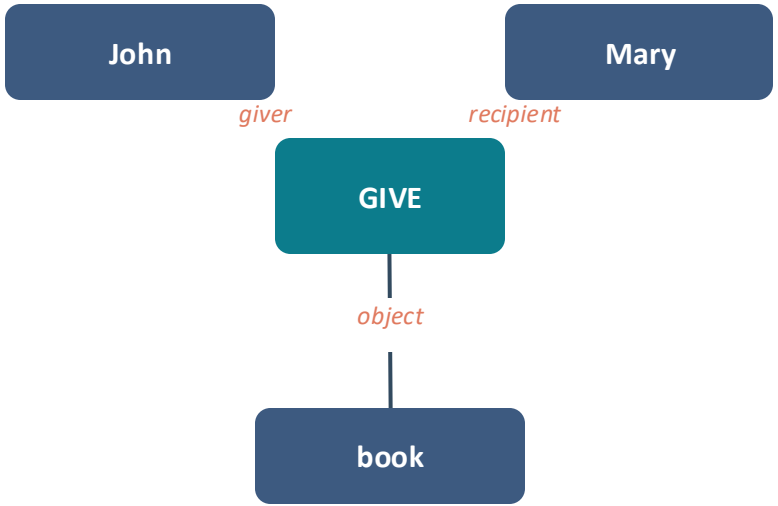
Nodes and Arcs: Binary Relations



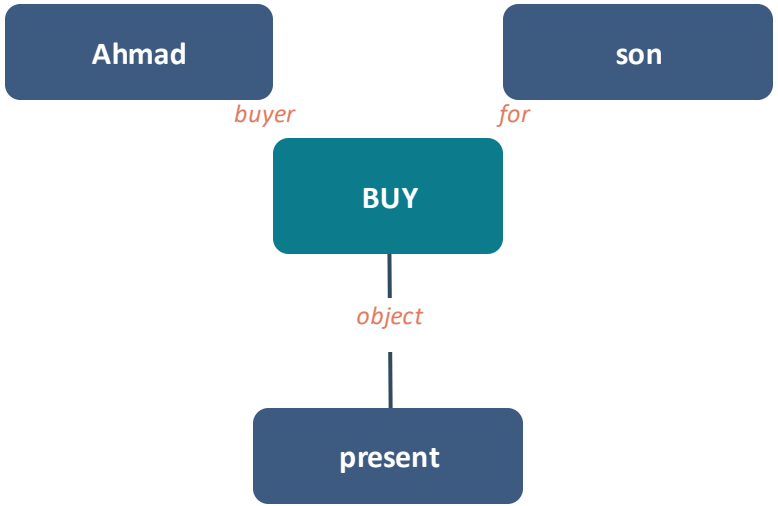
mother(john, sue)
wife(max, sue)
age(john, 5)
age(max, 34)

Non-binary Relations: Events

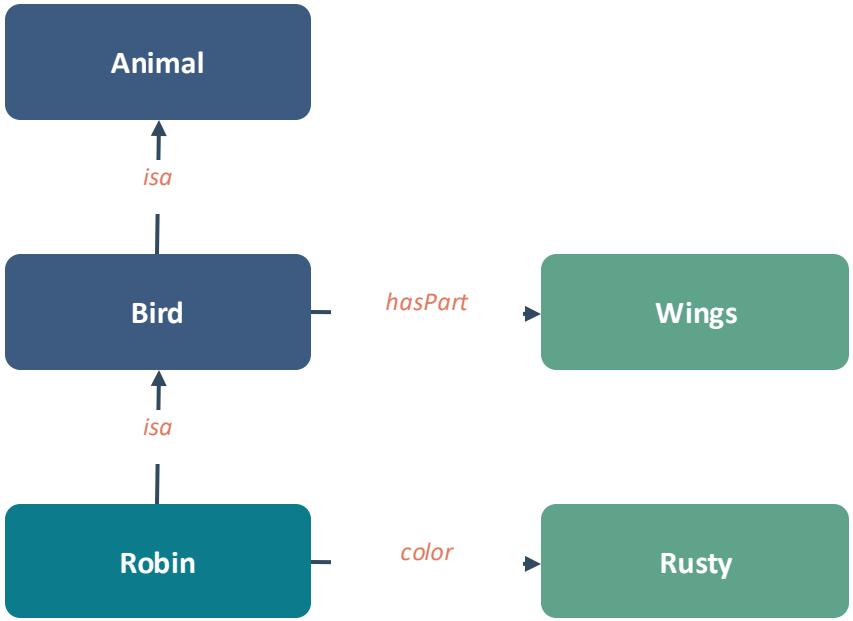
John gives Mary a book



Ahmad buys a present for his son

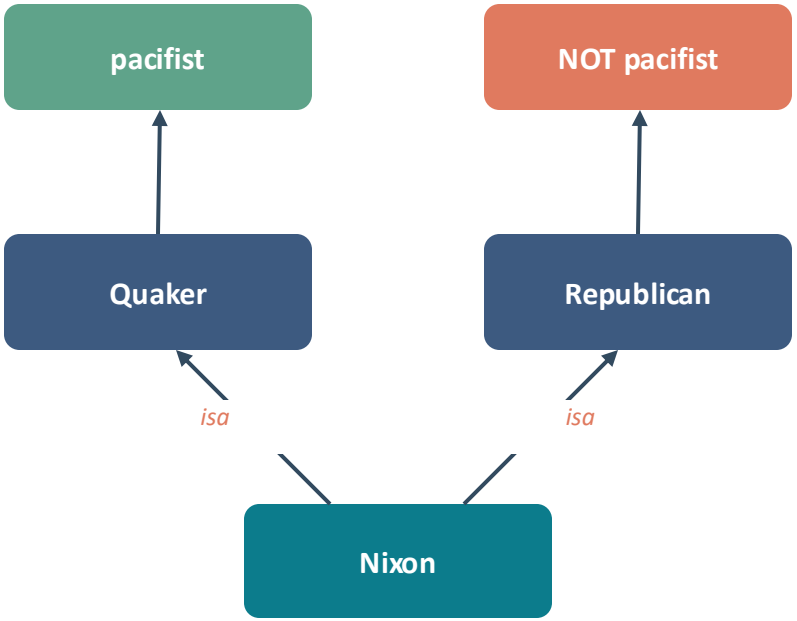


Inheritance (is-a)



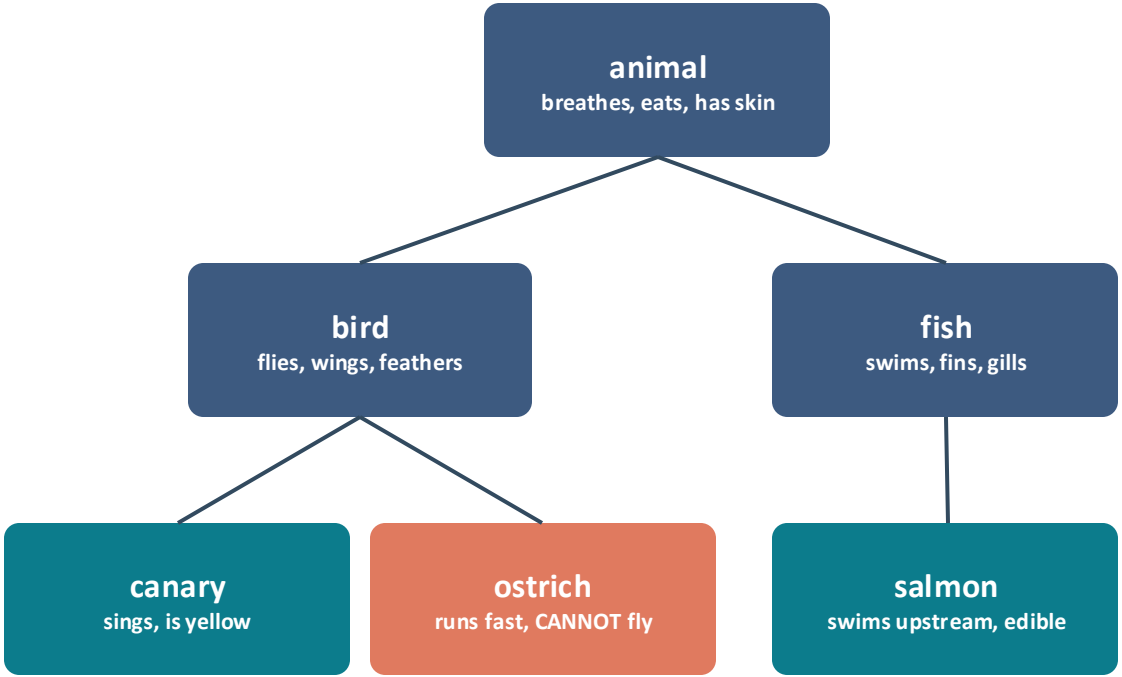
- isa links a class to its superclass.
- Some links (hasPart) are inherited along isa paths: a Robin has wings.
- Semantics may be informal or very formal, often fixed at implementation level.

Multiple Inheritance and Conflicts



- A node can have any number of superclasses and inherits from all of them and their ancestors.
- **Conflict:** Is Nixon a pacifist?
- **Needs a rule:** priority, specificity, or skepticism.

Example: Animal Taxonomy with Exceptions



- Arrows are is-a links.
- **Inheritance:** canary can fly.
- **Exception:** ostrich overrides "can fly".

Advantages of Semantic Networks

Usability

Easy to visualize.
Formal definitions have been developed.
Related knowledge is easily clustered.

Efficiency

Efficient in space.
Objects represented only once.
Relationships handled by pointers.

Disadvantages of Semantic Networks

Inheritance problems

Multiple sources and exceptions create conflicts.

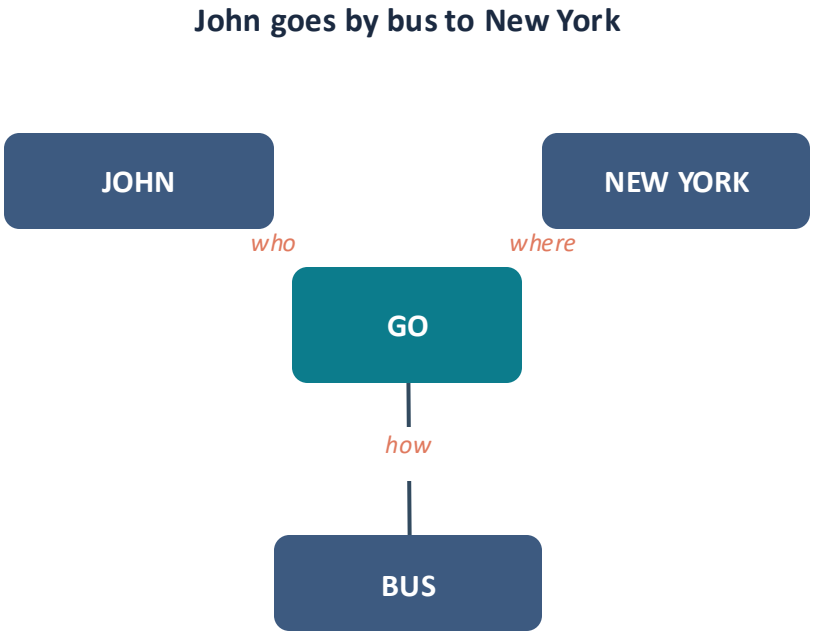
Misplaced facts

Facts placed at the wrong level cause wrong inferences.

No standards

No agreed standard for node and arc values.

Conceptual Graphs



- **Concept nodes:** individual (JOHN) or generic (BUS).
- **Relation nodes:** binary relations between concepts.
- Represents the meaning of simple natural-language sentences.

Frames

Frame: Book	
Slot	**Filler**
title	AI: A Modern Approach
author	Russell & Norvig
year	2003

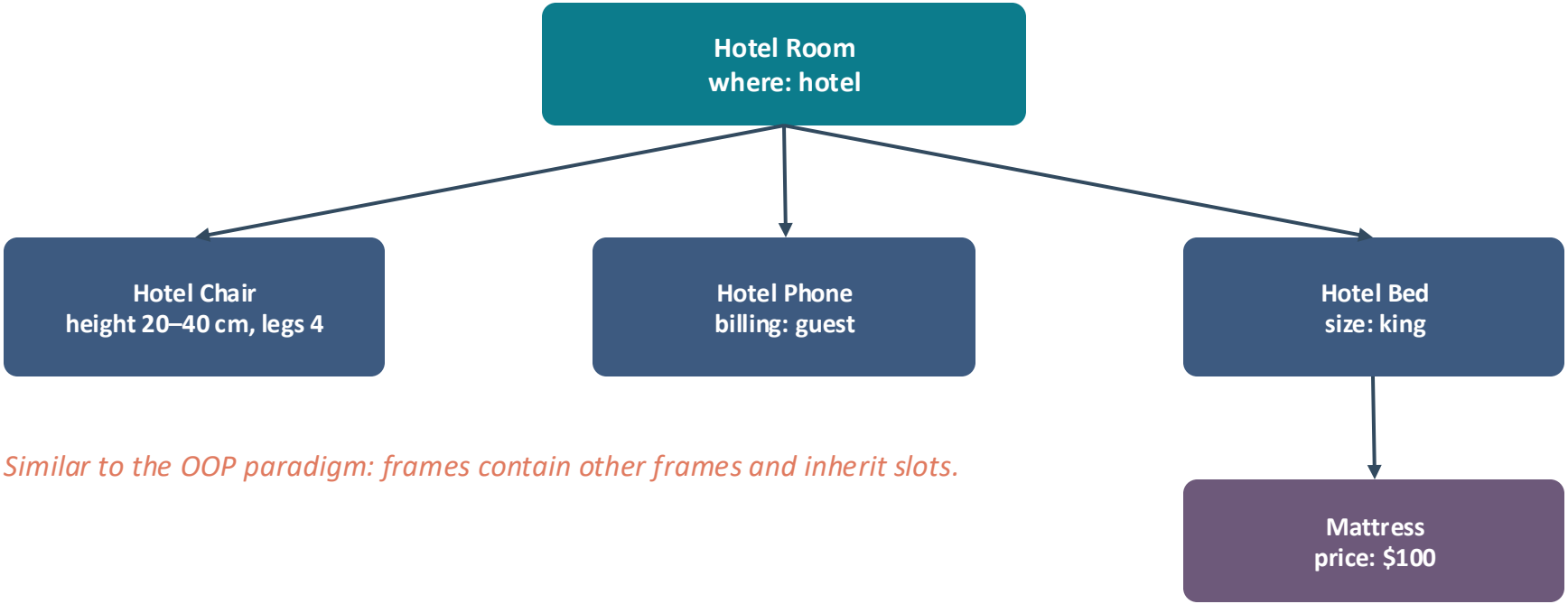
- A frame = set of slots (attributes) with values.
- Represents a specific entity or a general concept.
- A slot value can be another frame.
- **3 components:** frame name, slots, fillers (values, ranges, strings…).

Features of Frame Representation

- **Natural support of values:** each slot has constraints on legal values.
- Easy to implement with object-oriented techniques.
- Inheritance is easily controlled.

Frames	OOP
Frame	Class / object
Slot	Attribute
Filler	Value
Sub-frame	Subclass

Frame Inheritance: Hotel Example



Similar to the OOP paradigm: frames contain other frames and inherit slots.

Benefits of Frames

Organized

Groups related knowledge, making programming easier.

Accessible

Easily understood by non-developers.

Expressive & flexible

Expressive power; easy to add slots for new properties and relations.

Drawbacks of Frames

No standards

No standard for slot-filler values.

General methodology

A classroom frame differs for a professor and for a maintenance worker.

No inference

No associated reasoning or inference mechanism.

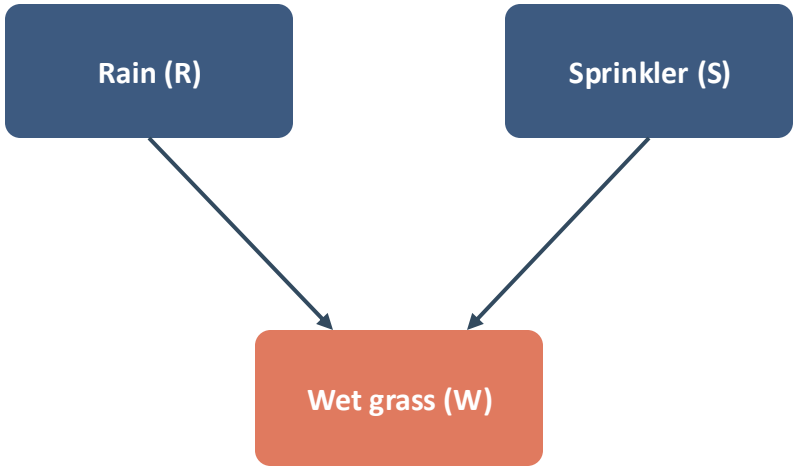
Method 2: Statistical Representation

Symbolic: certain facts
P = True / False

Statistical: likelihoods
P(event) between 0 and 1

- The real world is uncertain; symbolic logic fails when we cannot be 100% sure.
- **Core idea:** represent knowledge as likelihoods and distributions.
- Probabilistic graphs (e.g., Bayesian networks) model dependencies between variables.

Example: Why Is the Grass Wet?

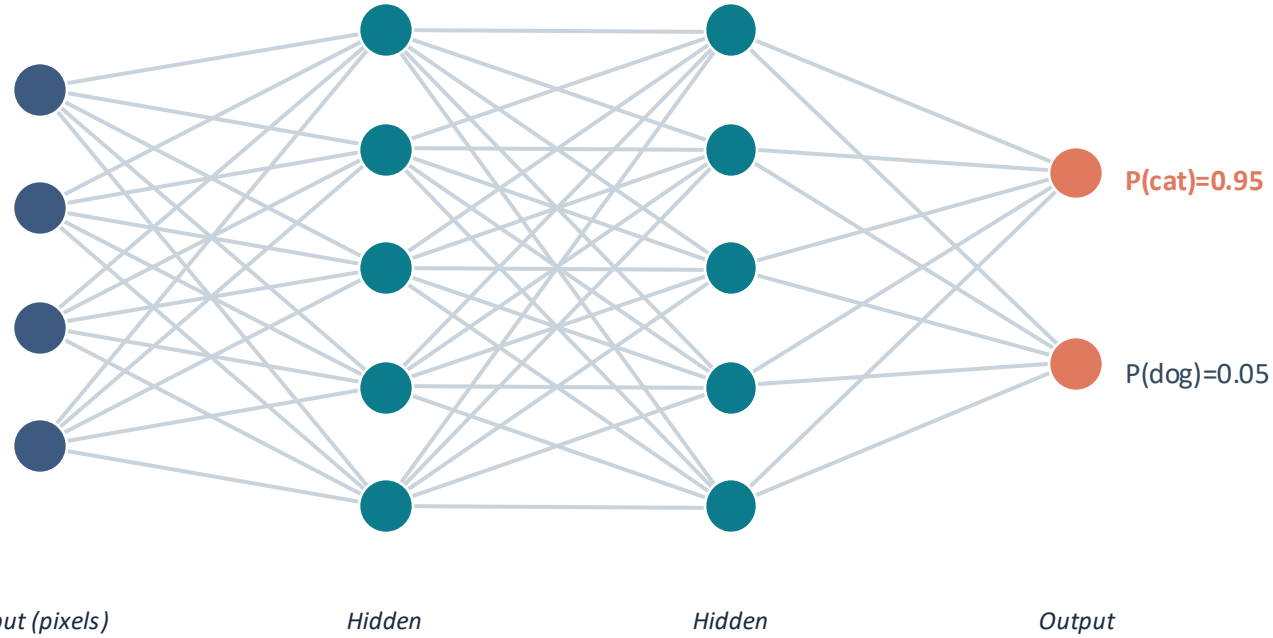


Probabilities (illustrative)

$P(R) = 0.2 \quad P(S) = 0.1$
$P(W|R,S) = 0.99$
$P(W|R,\neg S) = 0.90$
$P(W|\neg R,S) = 0.80$
$P(W|\neg R,\neg S) = 0$

- Observe W = True, then reason backwards: $P(R|W) \approx 0.74$ vs $P(S|W) \approx 0.34$.
- **Pros:** handles uncertainty, good for diagnosis. Cons: costly, needs data.

Method 3: Sub-symbolic Representation (Neural Networks)



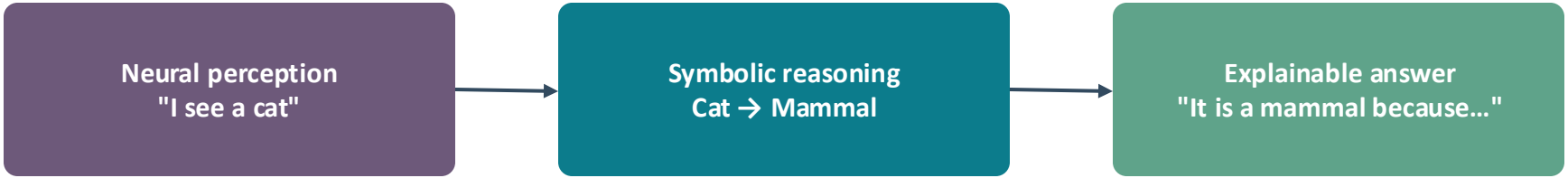
- Knowledge = weights, distributed across the network.
- **Pros:** perception, learns from data, robust to noise.
- **Cons:** black box; needs big data & compute.

Training adjusts millions of weights. No single neuron "means" cat; it is an emergent property.

A Practical Comparison: The "Cat" Problem

	Symbolic	Statistical	Sub-symbolic
Model	Rule: IF fur ∧ whiskers ∧ meow THEN Cat	P(Cat | Features)	Deep neural network
Input	Hand-written features	Feature probabilities, e.g. P(Cat|fur)=0.7	Pixels as numbers
Strength	Explainable, precise	Best guess with missing information	Learns its own features
Weakness	Fails if cat is silent or partly visible	Needs good data	Black box; data-hungry
Analogy	Librarian	Detective	Child learning by seeing

Conclusion: The Future Is Neuro-Symbolic



- **History:** from hand-designed symbolic to data-learned sub-symbolic representations.
- Symbolic AI = a librarian; statistical AI = a detective; sub-symbolic AI = a child learning by seeing.
- **Neuro-symbolic AI:** perceptual power + reasoning and explainability.

Practice Exercises

- **Draw a semantic network:** "Tweety is a canary; canaries are birds; birds fly." What does Tweety inherit?
- Design a frame for Student and a sub-frame PhD Student. List slots, fillers and inherited slots.
- Using the table on the Bayesian slide, compute P(W) and P(R | W).
- **Choose a paradigm for:** a spam filter, a rule-based medical expert system, face recognition. Justify.
- **Write in FOL:** "Every cat chases every mouse" and "Tom is a cat".

References

- Russell, S. & Norvig, P. Artificial Intelligence: A Modern Approach, 4th ed., Pearson, 2020.
- Brachman, R. & Levesque, H. Knowledge Representation and Reasoning, Morgan Kaufmann, 2004.
- Sowa, J. F. Knowledge Representation: Logical, Philosophical, and Computational Foundations, Brooks/Cole, 2000.
- Minsky, M. "A Framework for Representing Knowledge," MIT AI Lab Memo 306, 1974.
- Quillian, M. R. "Semantic Memory," in Minsky (ed.), Semantic Information Processing, MIT Press, 1968.
- Pearl, J. Probabilistic Reasoning in Intelligent Systems, Morgan Kaufmann, 1988.
- Goodfellow, I., Bengio, Y. & Courville, A. Deep Learning, MIT Press, 2016.
- Garcez, A. & Lamb, L. "Neurosymbolic AI: The 3rd Wave," arXiv:2012.05876, 2020.